JASON PLOWMAN

plowman.jason@gmail.com • +1-240-515-8830 • Germantown, MD

SUMMARY

Senior Machine Learning Engineer with 20+ years of software engineering experience and 5+ years building and deploying production ML systems in mission-critical environments. Expert in Python-based data pipelines, PyTorch modeling, and end-to-end ML lifecycle development from large-scale data ingestion through real-time inference and monitoring. Active Secret clearance with extensive experience delivering analytics and AI capabilities for government and defense applications.

CLEARANCE

- DoD **Secret Security Clearance**
- Eligible and willing to upgrade to **TS/SCI**
- Experienced working in **SCIFs**

EXPERIENCE

Mobius, ***Sr Software Engineer / Machine Learning Engineer***

2022 - 2026

Designed and deployed production machine learning systems for large-scale network telemetry analysis, enabling real-time traffic classification and operational insight.

- Architected **end-to-end ML data pipelines** for high-volume network traffic datasets, including ingestion, feature engineering, model training, validation, and production inference.
- Developed **PyTorch LSTM models** for real-time traffic classification, improving detection performance by **80% F1** while maintaining strict latency constraints.
- Optimized **real-time inference pipelines** for high-throughput environments, improving system responsiveness and scalability for operational deployments.
- Built and maintained **Apache NiFi ETL workflows and custom processors** to process and normalize large-scale telemetry data streams, improving processing efficiency by ~25%.
- Implemented **containerized ML deployment workflows** using Docker and CI/CD automation (GitHub Actions) to support reproducible model training, packaging, and deployment.
- Collaborated with engineering and operational stakeholders to translate mission requirements into ML-driven data products and analytics capabilities.
- Maintained full **ML lifecycle management**, including model evaluation, deployment monitoring, retraining pipelines, and operational performance tuning.
- Developed exploratory data analysis and visualization workflows to support model validation and operational decision-making.

Jacobs, ***Software and Systems Architect***

2013 - 2022

- Managed a small team (5 engineers) delivering secure, network-enabled telephony and communications platforms for DoD clients.
- Led platform updating effort to modernize infrastructure's OS, software packages, and languages.

- Applied Machine (Deep) Learning Neural Network models to invent novel, patentable solutions for client- (1) Threat Assessment from voiceprinting inbound callers, and (2) Call operator voice identity protection by natural sounding deep-fake voice masks.
- Managed up to 5 live voice and data communications platforms simultaneously.

Lockheed Martin, ***Software Consultant***

2012 - 2013

- Completed real-time software modules and updated interfaces in C++ for RTOS VxWorks.
- Automated testing using Perl scripts.

Thales, ***Sr. Embedded Software Engineer***

2010 - 2012

- Full software lifecycle development using CORBA, C++, Linux, and VxWorks
- Used Scrum Agile Methodology in my work (Daily Scrum Meeting, Planning Poker, Sprint Backlog, 1on1 meeting)

Renaissance Lighting, ***Embedded Systems Engineer***

2008 - 2010

- Developed dynamic color engine for lighting fixture using microcontroller Microchip PIC32 in C.

Northrop Grumman, ***Sr. Software Engineer***

2004 - 2007

- Real-time software development (C/C++, PSos, assembly for Freescale PPC, DSP) utilizing CMMI Level 5 development process.

Lockheed Martin, ***Software and Systems Engineer***

2000 - 2004

- Developed real-time C++ embedded software distributed over fiber-optic ethernet network.

EDUCATION

Georgia Institute of Technology

MS • ***Computer Science • Machine Learning track***

GPA 4.0 / 4.0

Johns Hopkins University

MS • ***Computer Engineering***

GPA 4.0 / 4.0

University of Minnesota BS • ***Physics***

GPA 3.6 / 4.0

CERTIFICATIONS

Artificial Intelligence Certificate

Stanford University • *expected Apr 2026*

SKILLS

Python Data Stack

- Python
- pandas
- NumPy
- scikit-learn
- PyTorch
- Matplotlib

Other Skills

C/C++ • SQL • feature engineering • MLOps • MLflow • Amazon Web Services (AWS) • Network Protocols